Lecture 6.2
Inefficiency of Random Access-Based Multiple Access Protocols

1. Overview

Medium Access Control (MAC) protocols determine how devices share a common communication medium. Among them, **random access-based MAC protocols**—such as ALOHA, CSMA, and IEEE 802.11 (WiFi)—enable decentralized coordination using randomized backoff mechanisms. While this decentralized approach avoids centralized control overhead, it suffers from **fundamental inefficiencies**, particularly under moderate to high network load, due to repeated collisions and idle channel times.

2. The Backoff Mechanism and Its Limitations

In CSMA/CA based MAC, each node selects a random number from a **contention window (CW)** and begins a countdown. The first node to reach zero transmits, while others wait. Although conceptually simple, this process leads to significant inefficiencies:

- **Idle Channel Time**: During the countdown, the channel remains idle, even though no transmission occurs. This results in underutilization of available bandwidth.
- **Collision Probability**: When two or more nodes pick the same backoff value, simultaneous transmission leads to **packet collisions**. These collisions cause retransmissions and an **exponential increase in CW**, further delaying future transmissions.

Quantitative Insights:

- In IEEE 802.11, with **CW = 16**, even with **3 nodes**, the probability of a collision is **~18%**.
- Studies ([4]) indicate **over 30% throughput reduction** in congested wireless LANs due to excessive backing off.
- At **higher data rates (e.g., 54 Mbps)**, shorter packet channel times exacerbate inefficiency since **fixed slot durations (e.g., 9 µs)** consume a greater fraction of the channel time.

3. Detailed Inefficiencies

3.1 Wasted Channel Time from Idle Slots

- **Slot duration** is lower bounded by propagation delay, clear channel assessment (CCA) time, and processing delays. For 802.11, this is typically **9 µs**.
- With high bitrate transmissions, e.g., **1500-byte packet at 54 Mbps**, the channel time is only **~222 µs**. If backoff involves, say, 8 slots on average, the idle time becomes **72 µs**, which is **nearly 25% of total channel time wasted before actual transmission**.

3.2 Exponential Backoff Under Congestion

- In IEEE 802.11, **Binary Exponential Backoff (BEB)** increases CW size as: `CW = 2^n * CWmin - 1`, for each failed attempt (up to CWmax = 1024).
- At **10 active nodes**, probability of at least two nodes choosing the same slot exceeds **40%**, drastically reducing throughput and increasing latency.

- Simulation and analytical results from [5], [6] show that throughput saturates and even **degrades** as network density increases beyond a threshold.

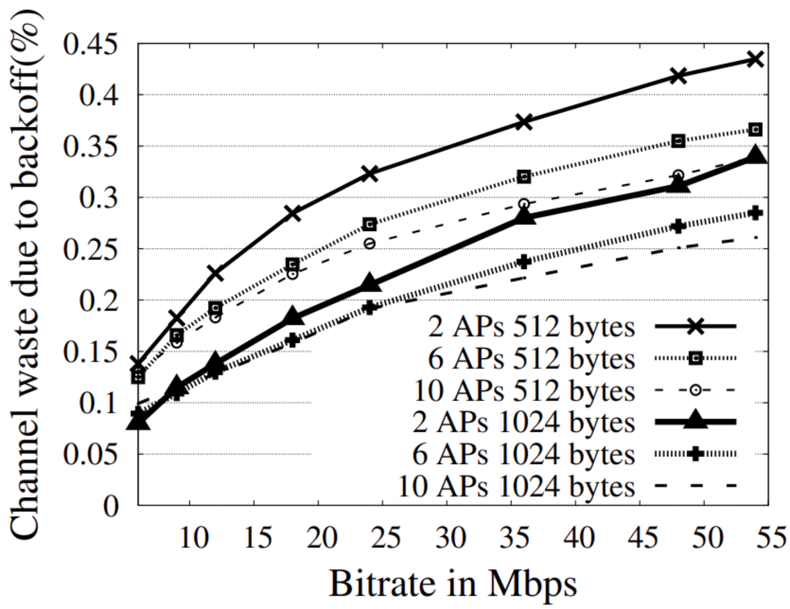

Fig. 1. Overhead of 802.11 backoff. Larger fraction of channel wasted with smaller packets at higher bitrates.

Figure 1 illustrates channel under-utilization across varying data rates and number of contending nodes. As node density increases, idle time and collision probability increase disproportionately, demonstrating a **non-linear degradation** of performance.

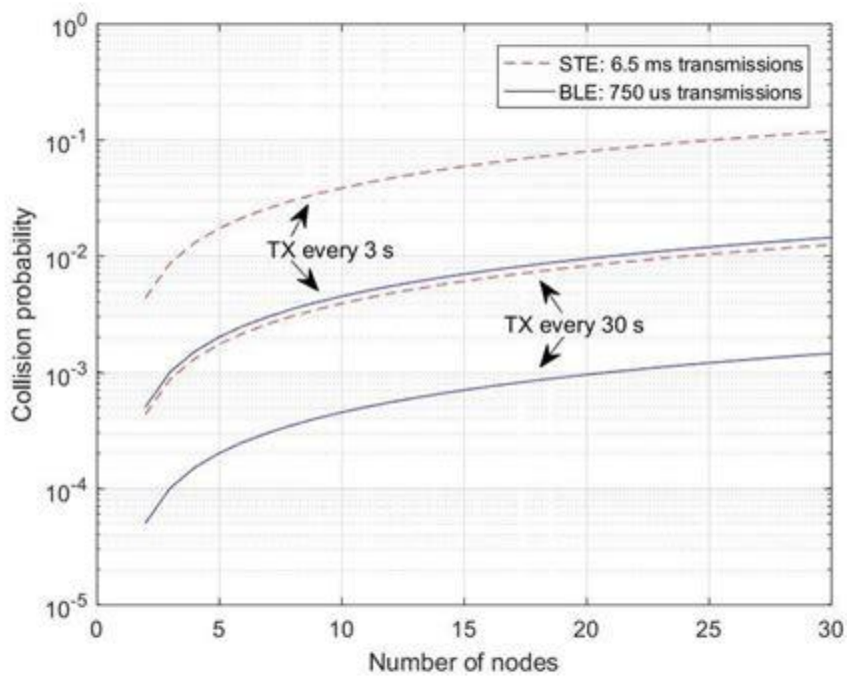

Figure 2: Collision Probability vs. Number of Nodes

This figure shows how collision probability increases nonlinearly with the number of contending nodes. Even with **10 nodes**, collisions exceed **40%**, leading to frequent retransmissions and reduced throughput. The inefficiency grows sharply with scale, limiting the practicality of random access MAC in dense networks.

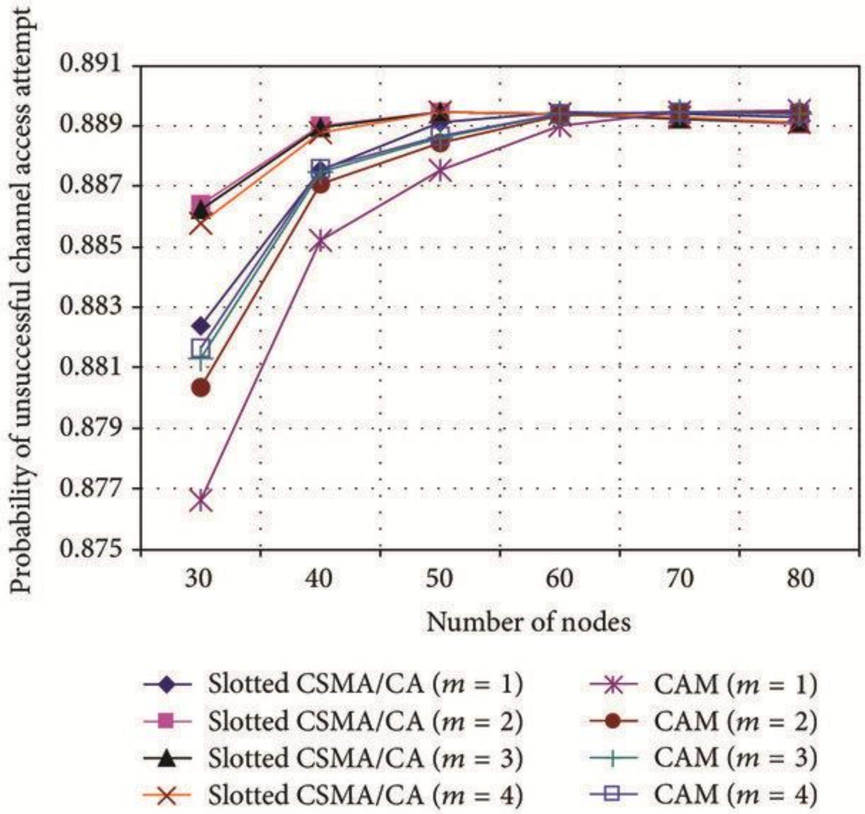

Figure 3: Collision Probability vs. Channel Load

As channel load increases, collision probability rises rapidly exceeding **75%** beyond a threshold (e.g., **1 pkt/sec**). This illustrates how CSMA/CA becomes highly inefficient under high traffic, highlighting the need for smarter or hybrid MAC strategies.

5. Implications and Future Directions

The drawbacks of random-access MAC highlight the need for:

- **Smarter contention resolution**, e.g., using machine learning or adaptive CW adjustment.
- **Hybrid protocols**, combining scheduled and contention-based approaches.
- **Resource reservation mechanisms**, such as **Semi-Random Backoff** ([3]), which aim to reduce collision probability.

References

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3. He, Y., Yuan, R., Sun, J., & Gong, W. (2009). *Semi-Random Backoff: Towards resource reservation for channel access in wireless LANs*. IEEE ICNP.
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